

Mobile Programming

Lecture 7

**Let's
discuss
Mortgage
Calculator**

Recall!

Quiz time !

Close your laptops and take away your phones from table and **ALL ATTENTION TO THE DESK**

1. Which language is commonly used for developing Android apps?

- A) Swift
- B) Kotlin
- C) Objective-C



2. What does mobile programming often involve?

- A) Only designing websites
- B) Only creating games
- C) Developing native and cross-platform apps

3. What is the purpose of `System.out.println("Hello World");` in Java?

- A) To save data to a file
- B) To display text on the screen
- C) To take input from the user

4. What symbol is used in Java to write a single-line comment?

- A) `/* */`
- B) `//`
- C) `#`

5. What is required if you want to swap the values of two variables in Java?

- A) Use if condition
- B) Use the ternary operator
- C) Use a third (temporary) variable

6. What is the result of this compound assignment: `k += 3`?

- A) It adds 3 to k
- B) It subtracts 3 from k
- C) It multiplies k by 3

7. What data type is used in Java to store a Unicode character?

- A) String
- B) char
- C) int

8. What is special about the String class in Java?

- A) It is mutable(changeable)
- B) It is a primitive data type
- C) It is immutable (unchangeable)

9. What does the `length()` method do in Java?

- A) Converts a string to lowercase
- B) Returns the number of characters in a string
- C) Compares two strings

10. Which Java class is used to take input from the user?

- A) System
- B) Scanner
- C) InputStream

A solid blue rectangular box, likely a placeholder for the correct answer to question 10.

11. What does (int)num do in Java?

- A) Declares a variable as integer
- B) Converts a variable to integer
- C) Compares a variable with integer

A solid blue rectangular box, likely a placeholder for the correct answer to question 11.

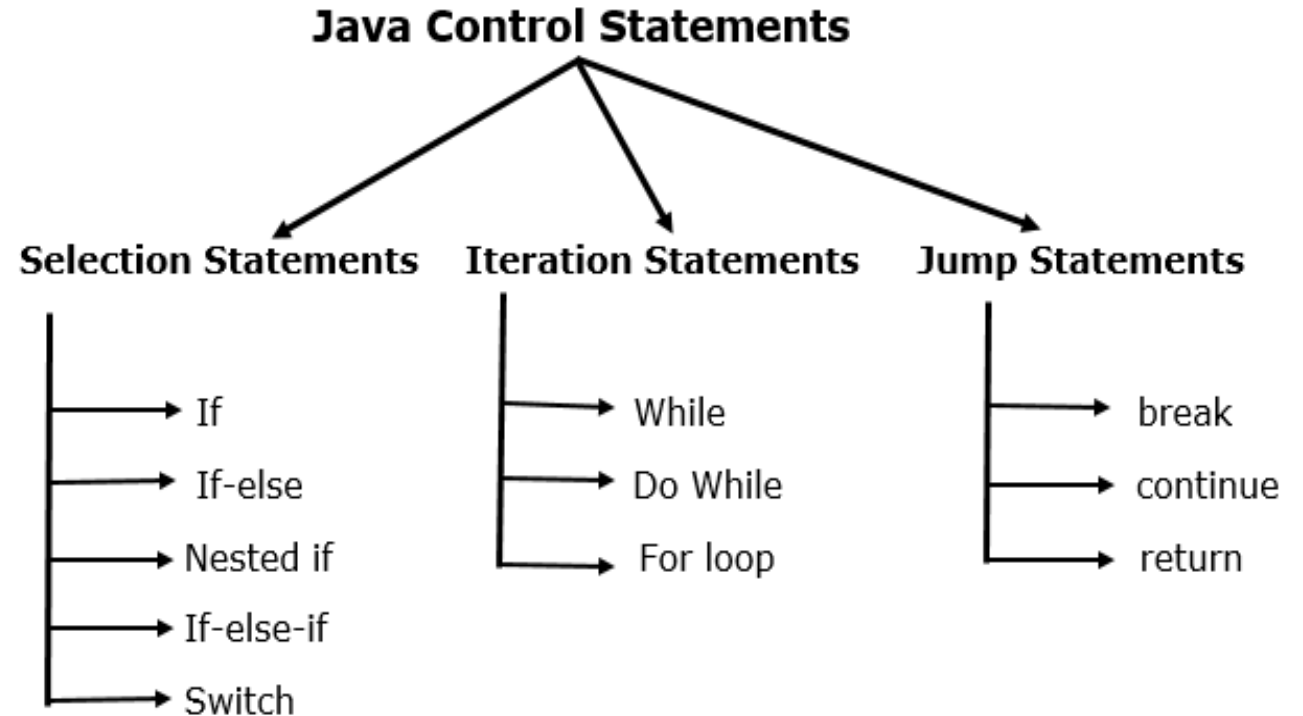
12. What is the purpose of pseudocode in programming?

- A) To replace code syntax
- B) To run a program
- C) To plan logic without worrying about syntax

A solid blue rectangular box, likely a placeholder for the correct answer to question 12.

Java - Decision Making

Decision making structures have one or more conditions to be evaluated or tested by the program, along with a statement or statements that are to be executed if the condition is determined to be true, and optionally, other statements to be executed if the condition is determined to be false.

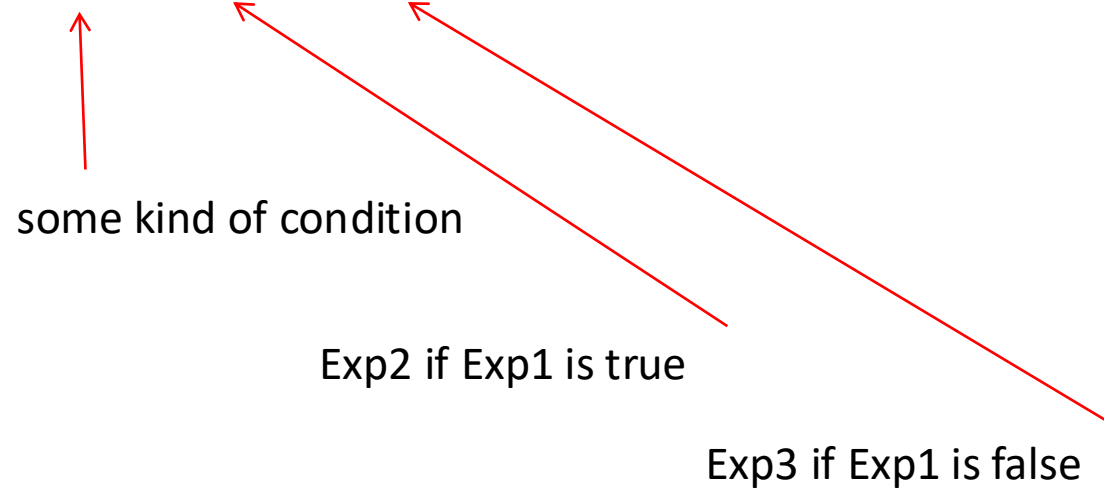


Sr.No.	Statement & Description
1	if statement An if statement consists of a boolean expression followed by one or more statements.
2	if...else statement An if statement can be followed by an optional else statement , which executes when the boolean expression is false.
3	nested if statement You can use one if or else if statement inside another if or else if statement(s).
4	switch statement A switch statement allows a variable to be tested for equality against a list of values.

The ? : Operator

The conditional operator ? : can be used to replace if...else statements. It has the following general form

Exp1 ? Exp2 : Exp3;



example:

THINK and try to find out the b for both cases

```
public class Test {  
  
    public static void main(String args[]) {  
        int a, b;  
        a = 10;  
        b = (a == 1) ? 20: 30;  
        System.out.println( "Value of b is : " + b );  
  
        b = (a == 10) ? 20: 30;  
        System.out.println( "Value of b is : " + b );  
    }  
}
```

If -else statement

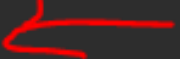
In Java, the if statement is a conditional statement used to execute a block of code when a specified condition evaluates to true. If the condition is false, an optional else statement can be used to execute an alternative block of code.

The if Statement

The if statement in Java checks a **Boolean expression** and executes a specific block of code only if the condition is **true**.

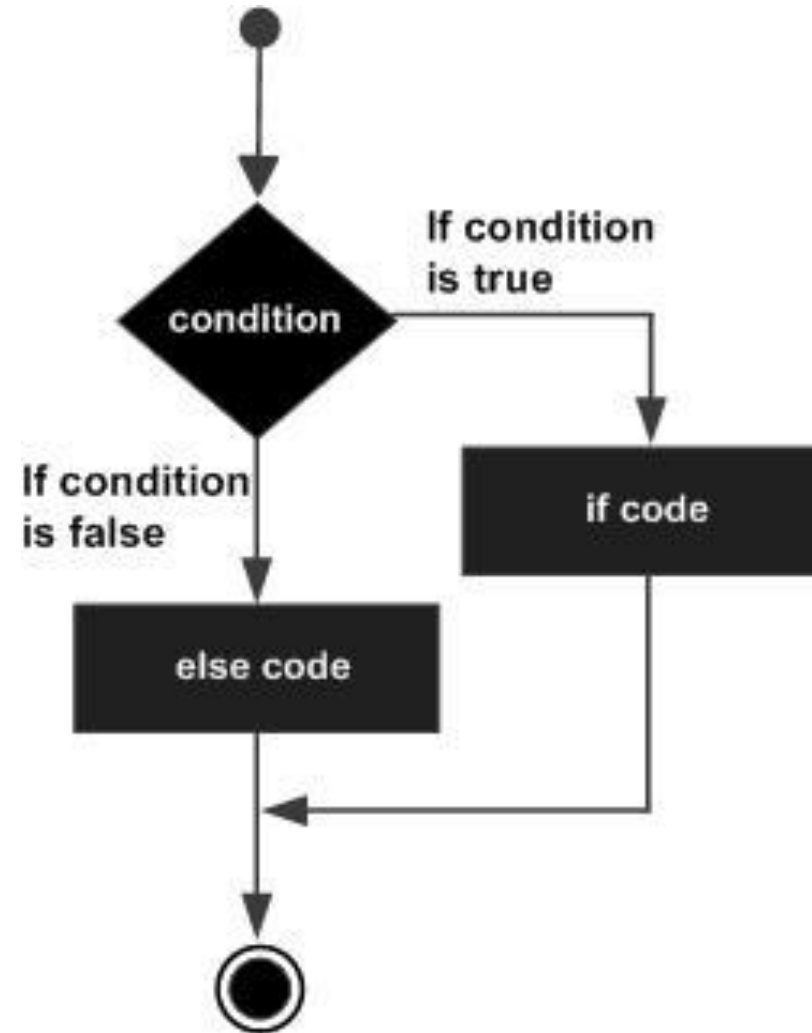
Syntax of if-else Statement

Following is the syntax of an if...else statement –

```
if(Boolean_expression) {   
    // Executes when the Boolean expression is true  
}else {   
    // Executes when the Boolean expression is false  
}
```

Flow Diagram

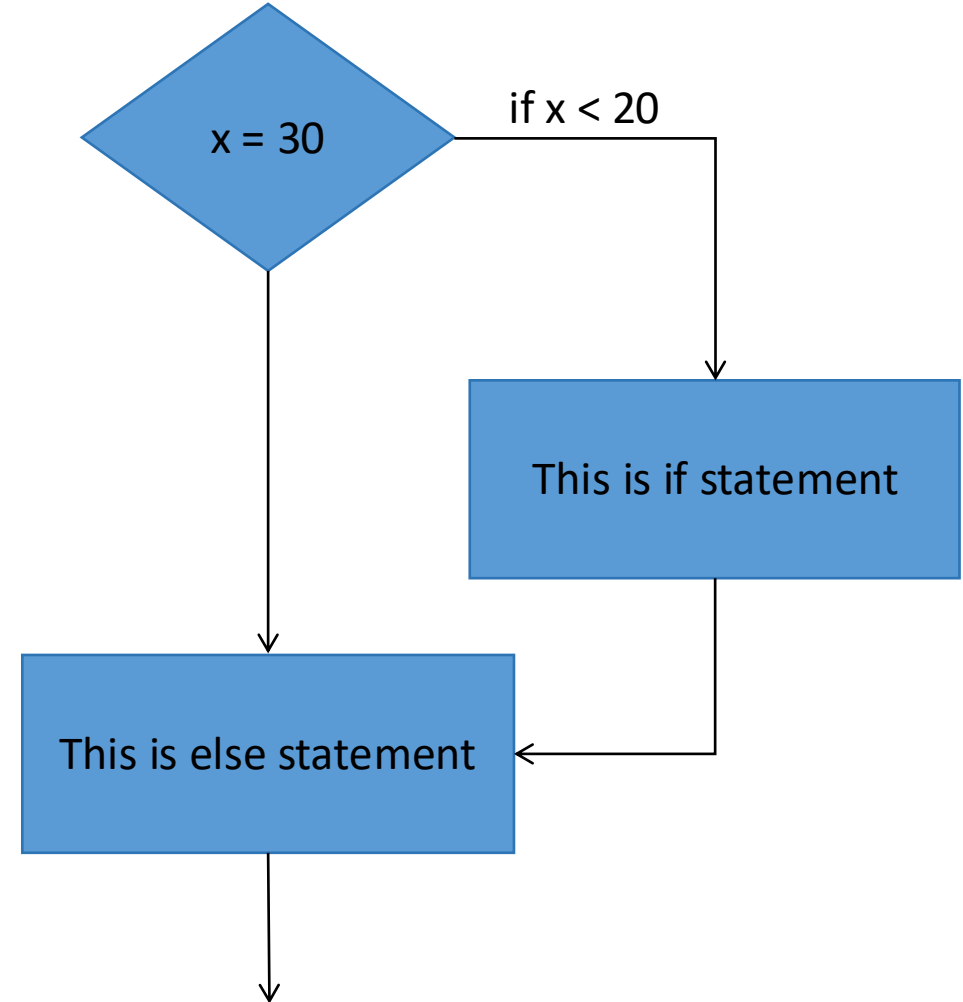
Below is the flowchart of the if-else statement, illustrating how it works:



Example of if else Statement

In this example, we're showing the usage of if else statement. We've created a variable x and initialized it to 30. Then in the if statement, we're checking x with 20. As if statement is false, the statement within the else block is executed.

```
public class Test {  
|  
    public static void main(String args[]) {  
        int x = 30;  
  
        if( x < 20 ) {  
            System.out.print("This is if statement");  
        }else {  
            System.out.print("This is else statement");  
        }  
    }  
}
```



Task 1: Write a Java program to get a number from the user and print whether it is positive or negative.

Test Data

Input number: 35

```
public class GachonUniversity{  
    //THE MPRTAGE CALCULATOR  
    Run | Debug  
    public static void main(String[] args){  
        Scanner obj = new Scanner(System.in);  
        int x = obj.nextInt();  
        if (x > 0){  
            System.out.println(x:"The number is positive");  
        }  
        else  
            System.out.println(x:"The number is negative");  
    }  
}
```

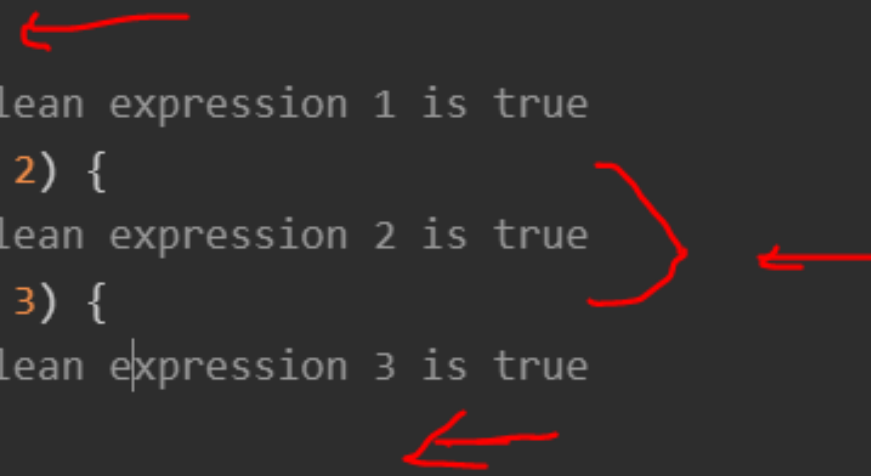
```
ding_f618cf5f\bin' 'GachonUniversity'  
3531fdf91d88a2411f\x5credhat.java\x5cjd_t_ws\x5c  
The number is positive
```

Task 2: Not try to give negative number to the same task

Syntax

Following is the syntax of an if...else if...else statement –

```
if(Boolean_expression 1) {  
    // Executes when the Boolean expression 1 is true  
}else if(Boolean_expression 2) {  
    // Executes when the Boolean expression 2 is true  
}else if(Boolean_expression 3) {  
    // Executes when the Boolean expression 3 is true  
}else {  
    // Executes when the none of the above condition is true.  
}
```

Hand-drawn red annotations on the code: an arrow points to the opening curly brace of the first 'if' block; a bracket groups the three 'else if' blocks, with an arrow pointing to the closing curly brace of the third 'else if' block; an arrow points to the 'else' block.

Example of if (if else if) Statements

In this example, we're showing the usage of if...else if...else statement. We've created a variable x and initialized it to 30. Then in the if statement, we're checking x with 10. As if statement is false, control jumps to else if statement checking another value with x and so on.

```
public class Test {  
  
    public static void main(String args[]) {  
        int x = 30;  
  
        if( x == 10 ) {  
            System.out.print("Value of x is 10");  
        }else if( x == 20 ) {  
            System.out.print("Value of x is 20");  
        }else if( x == 30 ) {  
            System.out.print("Value of x is 30");  
        }else {  
            System.out.print("This is else statement");  
        }  
    }  
}
```

THINK and try to find out the answer!

The answer is :

Value of X is 30

Task 3: Find Greatest Among Three Numbers

Write a Java program that takes three numbers from the user and prints the greatest number.

Test Data

Input the 1st number: 25

Input the 2nd number: 78

Input the 3rd number: 87

```

5 public class GachonUniversity{
6     //THE MPRTAGE CALCULATOR
    Run | Debug
7     public static void main(String[] args){
8         Scanner obj = new Scanner(System.in);
9         int num1 = 25;
10        int num2 = 78;
11        int num3 = 87;
12        if (num1 > num2 && num1 > num3){
13            System.out.println("The greteast number is :" + num1);
14        }
15        else if (num2 > num1 && num2 > num3){
16            System.out.println("The greteast number is :" + num2);
17        }
18        else
19            System.out.println("The greteast number is :" + num3);
20    }
21

```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

```

● ionMessages' '-cp' 'C:\Users\Shabir\AppData\Roaming\Code\User\workspaceStorage\a052f342cec4
ding_f618cf5f\bin' 'GachonUniversity'
The greteast number is :87

```



In my code i directly made input
not user input
But in your must be user input

Task 4: Check Positive, Negative, or Zero (with Range)

Write a Java program that reads a floating-point number and prints "zero" if the number is zero. Otherwise, print "positive" or "negative".

Test Data

you have to check for all conditions:

Input a number: 25 #for positive condition

Input a number: 0 #for zero condition

Input a number: -25 #for negative condition

```
public class GachonUniversity{  
    //THE MPRTAGE CALCULATOR  
    Run | Debug  
    public static void main(String[] args){  
        Scanner obj = new Scanner(System.in);  
        int num1 = 25;  
  
        if (num1 == 0){  
            System.out.println(x:"The number is zero");  
        }  
        else if (num1 > 0){  
            System.out.println(x:"The number is positive");  
        }  
        else  
            System.out.println(x:"The number is negative");  
    }  
}
```